

GMO IM.E.K. FEDOROVA, Russian Federation

WMO: 202920

Lat: 77.713N

Lon: 104.292E

Elev: 49

StdP: 14.67

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-39.3	-35.1	-42.5	0.5	-38.9	-38.7	0.6	-34.8	42.1	0.7	37.4	-3.3	7.7	250	1.200

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.3	48.1	46.0	43.8	42.2	40.3	39.0	46.1	47.9	42.3	43.6	39.1	40.2	11.8	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
44.7	43.7	47.0	40.9	37.7	43.3	37.9	33.5	39.7	18.2	48.0	16.3	43.9	14.8	40.3	63.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 34.5	30.5	27.4		-39.7	55.8	6.4	6.0	-44.3	60.1	-48.0	63.6	-51.6	66.9	-56.2	71.3	
(5)			DB	-39.8	51.6	6.8	4.5	-44.8	54.8	-48.8	57.5	-52.6	60.0	-57.6	63.3	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.6	-14.9	-16.0	-13.1	-1.3	14.9	30.5	34.7	35.0	29.4	15.4	-1.8	-10.9	
	DBStd	21.60	11.71	11.37	12.19	10.69	8.36	4.13	3.66	4.42	5.74	10.72	10.16	10.96	
	HDD50	15100	2011	1847	1957	1540	1087	586	474	466	618	1073	1553	1888	
	HDD65	20574	2476	2267	2422	1990	1552	1036	938	931	1068	1538	2003	2353	
	CDD50	2	0	0	0	0	0	0	1	1	0	0	0	0	
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0	
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0	
Wind	WSAvg	13.1	13.7	14.0	12.3	12.9	12.5	13.1	12.9	12.9	12.8	13.7	13.3	13.2	
Precipitation	PrecAvg	8.6	0.5	0.5	0.6	0.4	0.5	0.7	1.2	1.3	0.9	0.8	0.5	0.6	
	PrecMax	14.3	2.3	2.0	2.6	1.4	2.0	2.0	3.0	3.8	2.2	2.2	1.5	2.1	
	PrecMin	3.5	0.0	0.1	0.0	0.0	0.1	0.1	0.1	0.2	0.3	0.1	0.1	0.1	
	PrecStd	2.4	0.4	0.4	0.5	0.3	0.4	0.5	0.8	0.8	0.5	0.4	0.3	0.4	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	23.5	18.3	24.2	31.8	34.8	45.1	56.1	55.6	46.2	36.1	29.4	24.5	
		MCWB	22.6	17.9	23.5	30.9	33.3	42.9	52.2	51.9	44.7	34.8	28.4	23.9	
	2%	DB	14.5	10.6	18.9	24.7	31.8	39.3	48.0	49.1	42.5	33.2	24.7	17.2	
		MCWB	13.8	10.2	18.3	23.6	30.9	37.0	45.6	47.1	41.3	32.4	23.9	16.6	
	5%	DB	6.8	5.1	11.7	18.8	29.1	36.8	42.7	44.7	39.2	31.0	16.7	9.1	
		MCWB	6.4	4.6	11.1	17.9	28.1	35.2	40.8	43.2	38.4	30.0	15.9	8.5	
	10%	DB	0.5	0.0	4.7	13.9	26.8	34.8	38.9	40.8	36.0	28.9	12.2	3.8	
		MCWB	0.0	-0.5	4.1	13.1	25.9	33.5	37.4	39.7	35.3	28.0	11.5	3.3	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	23.2	17.9	23.7	30.8	33.6	43.4	52.2	52.1	45.1	34.9	28.5	23.8
			MCDB	23.3	18.3	24.1	31.7	34.7	45.3	55.3	54.6	46.3	35.6	29.2	24.4
2%		WB	13.9	10.2	18.2	23.6	30.9	37.3	45.6	47.3	41.3	32.4	23.9	16.8	
		MCDB	14.3	10.5	18.8	24.6	31.7	38.7	47.9	48.8	42.2	33.1	24.7	17.3	
5%		WB	6.4	4.6	11.2	17.9	28.3	35.4	40.9	43.3	38.3	30.2	16.0	8.6	
		MCDB	6.8	5.0	11.8	18.7	28.9	36.7	42.6	44.7	39.1	30.9	16.6	9.1	
10%		WB	0.1	-0.5	4.1	13.1	25.9	33.8	37.5	39.7	35.5	28.2	11.5	3.4	
		MCDB	0.5	-0.1	4.6	13.8	26.7	34.5	38.6	40.7	35.9	28.9	12.2	3.8	
Mean Daily Temperature Range		5% DB	MDBR	10.1	9.6	10.2	10.1	7.3	4.6	5.0	5.3	4.7	8.1	9.7	9.7
			MCDBR	16.1	14.9	13.6	12.5	7.0	7.7	13.8	13.3	7.3	6.7	12.1	15.3
	MCWBR		16.0	14.8	13.7	12.2	6.8	6.3	11.7	11.2	6.6	6.7	12.1	15.1	
	5% WB	MCDBR	16.0	14.9	13.7	12.5	7.1	7.1	13.7	13.1	7.0	6.0	12.1	15.4	
		MCWBR	16.0	14.8	13.8	12.2	7.0	6.2	11.8	11.3	6.5	6.2	12.1	15.2	
Clear-Sky Solar Irradiance	taub	N/A	0.172	0.255	0.294	0.322	0.312	0.349	0.326	0.266	0.182	N/A	N/A		
	taud	N/A	1.873	1.975	2.042	2.058	2.261	2.305	2.397	2.293	1.937	N/A	N/A		
	Ebn at Noon	0	20	172	235	251	264	245	231	186	12	0	0		
	Edh at Noon	0	2	18	30	36	31	29	22	14	1	0	0		
All-Sky Solar Radiation	RadAvg	0	10	305	1106	1820	1926	1526	854	302	41	0	0		
	RadStd	0	1	31	59	88	177	97	91	31	6	0	0		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page